

CHEM 2312 LS

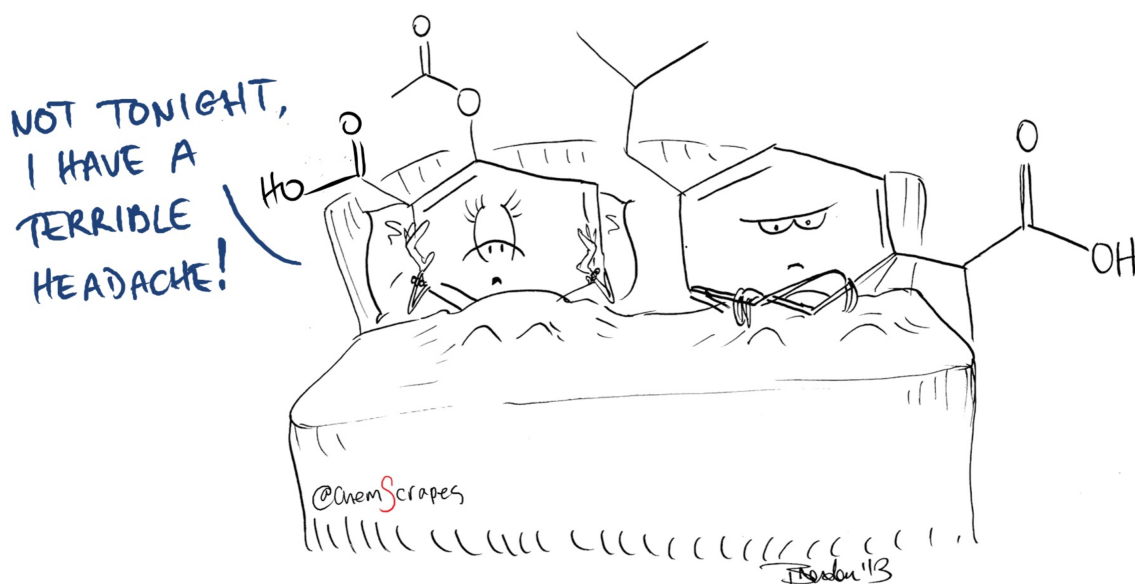
Organic Chemistry II

Midterm Examination 1

Take a deep breath. You got this.

Do not open this exam until you are instructed to do so. Only a pencil or pen, scratch paper, and molecular model kit are permitted. You will have 50 minutes to complete the exam; it can be submitted at any time after 25 minutes have elapsed. Write all responses clearly in the boxes or spaces provided; work outside of designated areas will not be graded. On multiple-choice questions, write the letter of your choice in the box *and* circle your choice.

Ensure that configurations can be unambiguously assigned at all stereocenters (*R/S* or *E/Z*). Use “+ En” to indicate the presence of a racemic mixture when chiral products are generated from achiral starting materials.



Name

GT ID

Multiple-choice Questions. Circle your choice and write its letter in the box.

Each question is worth **5 points**.

1. Which of the following monosubstituted benzenes reacts most *slowly* with methyl chloride (MeCl) and aluminium trichloride (AlCl_3)?

- A. PhNO_2
- B. PhCOMe
- C. PhMe
- D. PhOEt
- E. PhOMe

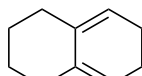
2. Which of the following compounds does not react as a diene in Diels-Alder reactions?



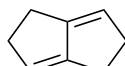
1



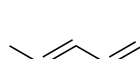
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3



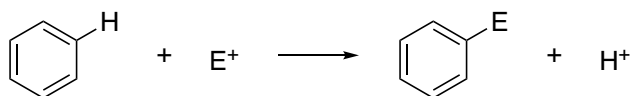
4



5

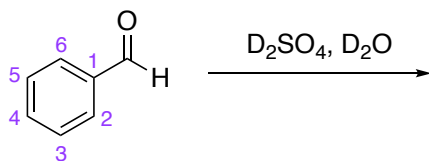
- A. **1**
- B. **2**
- C. **3**
- D. **4**
- E. **5**

3. Which of the following sequences of elementary steps best describes the general mechanism of electrophilic aromatic substitution, a reaction scheme for which is shown below?



- A. 1. $\text{S}_{\text{N}}2$; 2. A_{E}
- B. 1. A_{E} ; 2. pt
- C. 1. A_{N} ; 2. D_{N}
- D. 1. pt; 2. Ad_{N}
- E. 1. A_{E} ; 2. $\text{S}_{\text{N}}2$

4. When the compound below is treated with D_2SO_4 in D_2O , deuterium (D) is incorporated into the aromatic ring. At which carbon(s) does deuterium appear?


☐

- A. Carbon 1 only
B. Carbon 2 only
C. Carbon 4 only
D. Carbons 2, 4, and 6
E. Carbons 3 and 5

5. Which of the following is *not* a limitation of Friedel-Crafts alkylation?

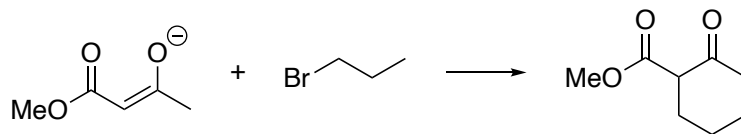
☐

- A. The active electrophile can rearrange after it is formed.
B. Electron-rich benzenes react at prohibitively slow rates.
C. Multiple alkyl groups are often added to the aromatic substrate in an uncontrolled manner.
D. Anilines ($ArNH_2$) do not yield products of electrophilic substitution.
E. Primary alkyl halides are unreactive under the reaction conditions.

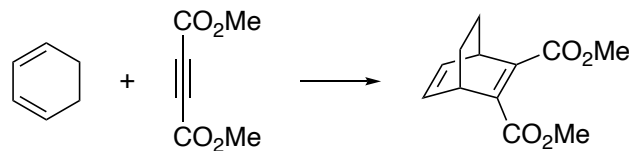
Other Questions. Place your response in the designated space for each question.

6. **6 pts.** Each of the transformations below occurs in a single elementary step. Add curved arrows and any necessary implied hydrogens to the reactants below to show the flow of electrons in each step. Lone pairs are optional.

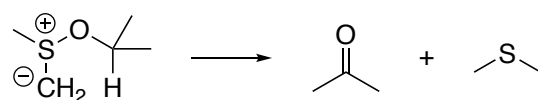
A.



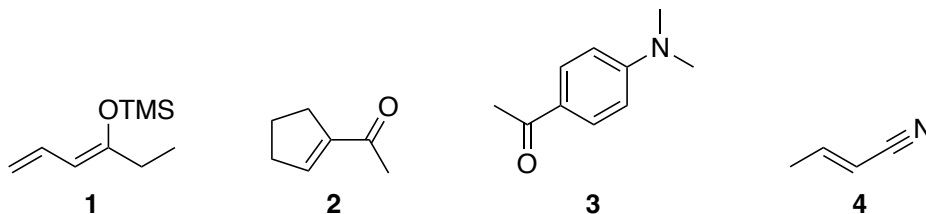
B.



C.



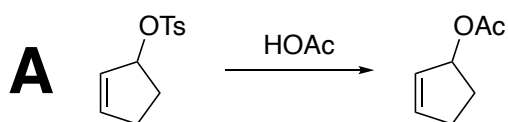
7. **10 pts.** Circle the electron-withdrawing groups in the structures below and draw a box around the electron-donating groups.



8. **6 pts.** Of the structures above, which has the *longest* wavelength of maximum absorbance λ_{max} ? Explain how you know.

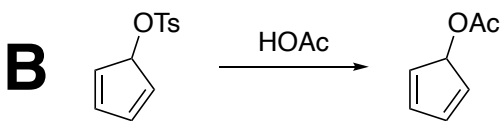
longest λ_{max} :

9. **13 pts.** Rank S_N1 reactions **A** – **C** from fastest to slowest. Provide the structure of the cationic intermediate in each reaction and identify it as aromatic, antiaromatic, or non-aromatic.



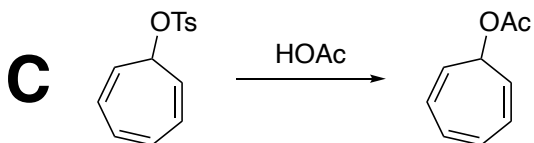
Intermediate **A**

- ☐ Aromatic
☐ Antiaromatic
☐ Non-aromatic



Intermediate **B**

- ☐ Aromatic
☐ Antiaromatic
☐ Non-aromatic



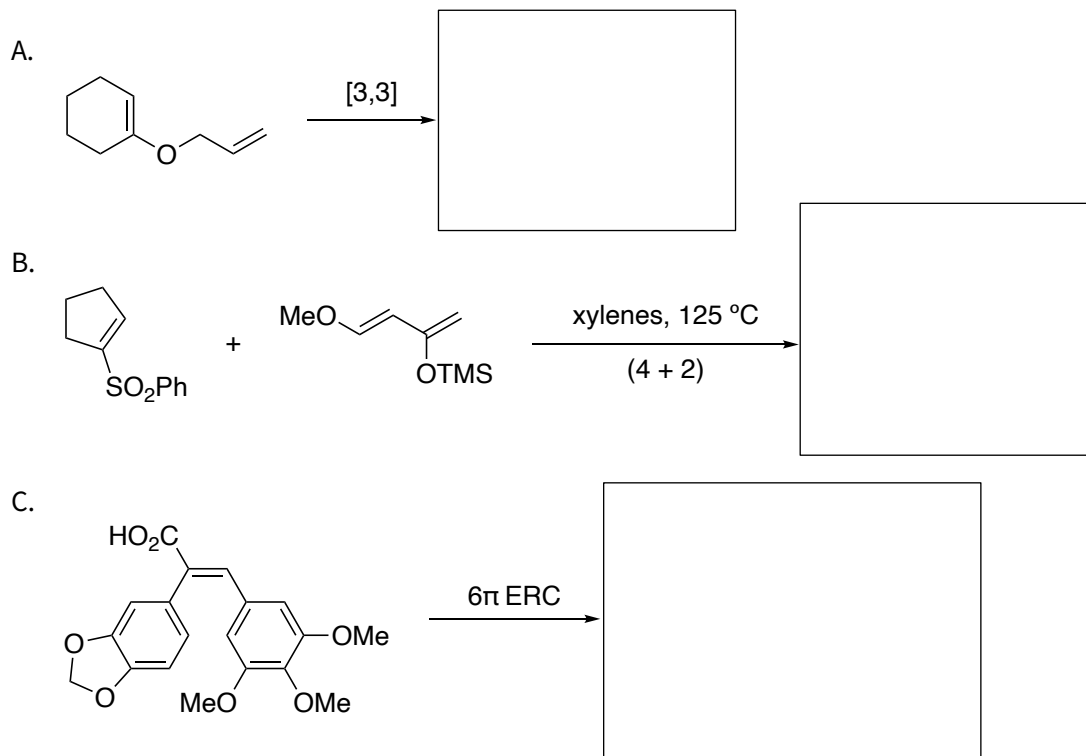
Intermediate **C**

- ☐ Aromatic
☐ Antiaromatic
☐ Non-aromatic

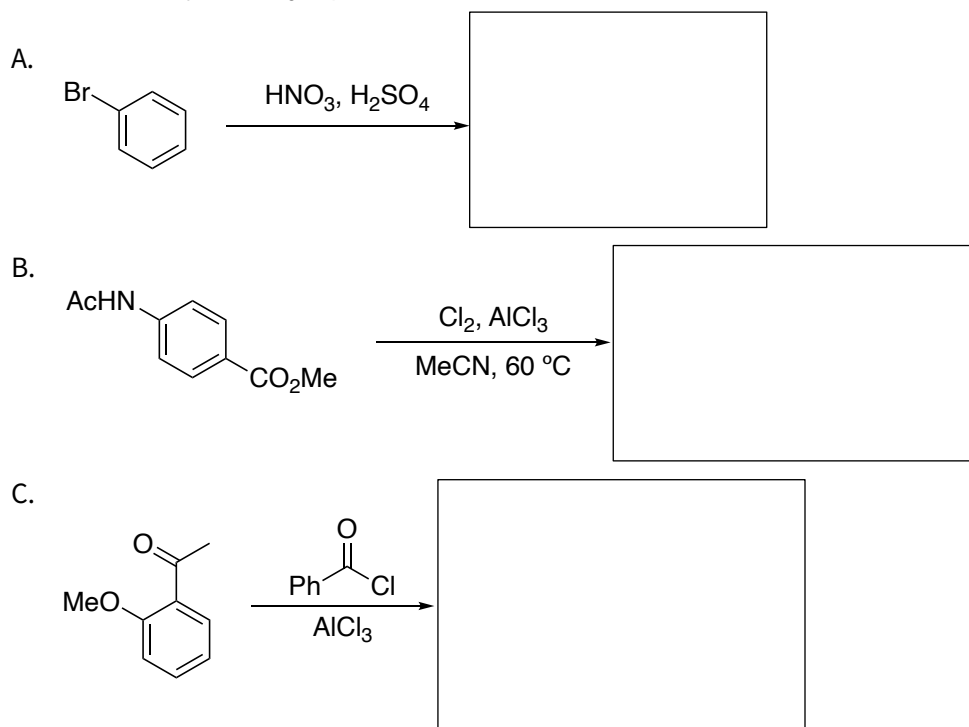
fastest

slowest

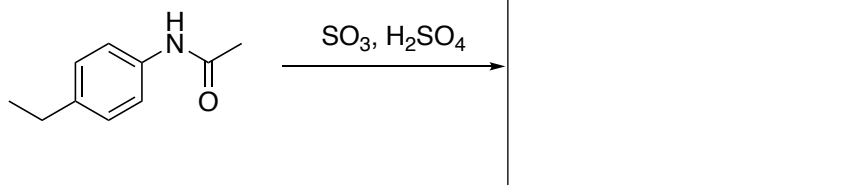
10. **12 pts.** Predict the neutral organic product of each reaction below.



11. **16 pts.** Predict the neutral organic product of each reaction below. If multiple isomers are possible, draw only the major product.

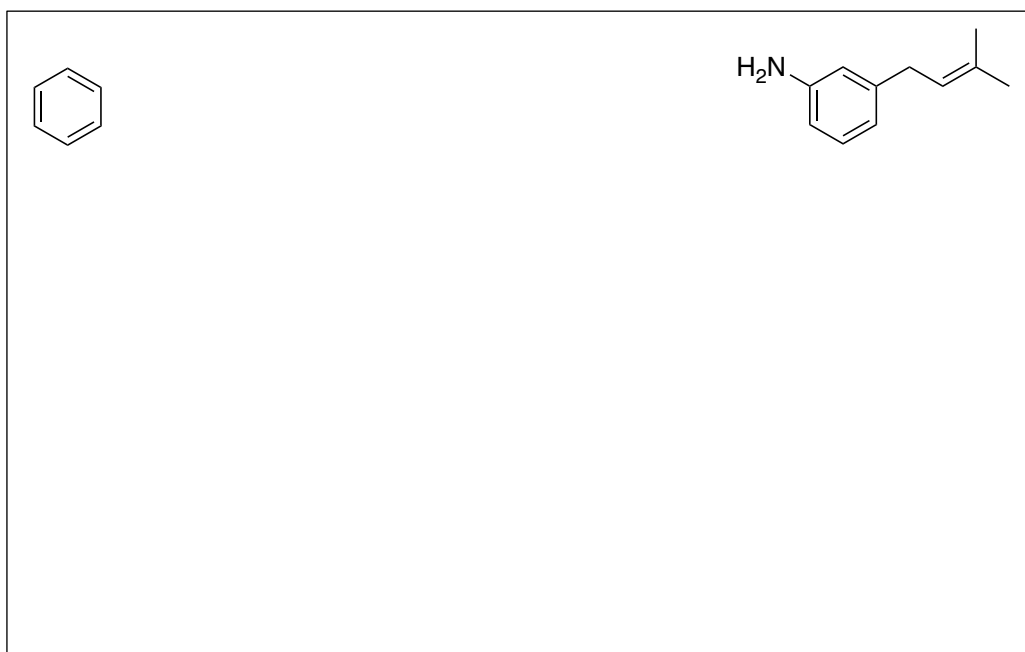


D.



12. **12 pts.** Show how each of the compounds below can be synthesized from benzene using a sequence of reactions. Include all reagents and structures for all isolated products in a synthetic scheme. Do not include mechanisms or reactive intermediates.

A.



B.

